AP CALCULUS AB – UNIT 4

4.3 Other Rates of Change

Name: ____________ **Class:** ____________ **Date:** ____________

Lesson Overview and Learning Objectives

- Calculate and interpret derivatives in scientific, economic, and geometric contexts.
- Distinguish a modeled quantity from its instantaneous rate of change.
- Use correct units and signs in applied interpretations.
- Combine derivative rules to calculate rates for derived quantities.

Keys: Essential Knowledge and Vocabulary

Rate in context	For $Q = Q(t)$, $Q'(t)$ describes output change per unit input.
Marginal quantity	For cost, revenue, or profit as a function of production, a derivative approximates change caused by one additional unit.
Sensitivity	If $Q = Q(x)$, then $Q'(a)$ measures how sensitive Q is to a small change in x near a.
Relative rate	$\frac{Q'(t)}{Q(t)}$ gives fractional change per input unit; multiply by 100% for percent rate.
Derived rate	Differentiate the formula defining a derived quantity, then substitute the required input.
Units check	Use dimensional analysis to detect impossible interpretations or algebraic errors.

Explanation, Strategy, and Common Errors

- Begin by naming the independent and dependent variables and their units.
- A derivative can be computed symbolically, estimated from data, or read as slope from a graph; the interpretation format remains the same.
- Do not confuse a large amount with a large rate. Compare rates using derivative values and magnitudes.
- For marginal interpretations, say 'approximately' because a derivative predicts a small discrete change.
- When a new quantity is built from several changing quantities, apply the appropriate product, quotient, or Chain Rule.

Solved Example: Marginal profit

Profit is $P(q) = 70q - 0.04q^2 - 5000$ dollars. Find and interpret $P'(600)$.

Answer and Reasoning

$P'(q) = 70 - 0.08q$, so $P'(600) = 22$. Near 600 units, one additional unit increases profit by approximately \$22.

Solved Example: Geometric sensitivity

The area of a circle is $A(r) = \pi r^2$. Find and interpret $A'(10)$.

Answer and Reasoning

$A'(r) = 2\pi r$, so $A'(10) = 20\pi$ cm^2/cm. Near radius 10 cm, area increases by about 20π cm^2 per additional centimeter of radius.

Solved Example: Relative growth

A culture satisfies $N(t) = 800e^{0.12t}$. Find its relative growth rate.

Answer and Reasoning

$N'/N = 0.12$, so the instantaneous relative growth rate is 12% per time unit.

Solved Example: Rate from a table

Using the shared table, interpret $W'(6) = 0.80$.

Answer and Reasoning

At hour 6, electrical power usage is increasing at 0.80 kilowatt per hour.

Shared Representation / Data

t (hours)	0	2	4	6	8
$W(t)$ (kW)	18.0	22.4	27.1	31.0	33.2
$W'(t)$ (kW/hour)	2.40	2.05	1.55	0.80	0.15

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For LH́opital problems, verify an allowed indeterminate form first.

Leveled Practice

Easy Foundations

Build fluency with the definition, notation, and one-step applications.

1. The radius r is in cm and area A in cm^2. State units of dA/dr.

2. Interpret $C'(250) = 8.60$ when C is cost in dollars and q is items.

3. If $T'(3) = -1.4$, is temperature increasing or decreasing?

4. Find $A'(x)$ if $A = x^2 + 4x$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Volume is $V(r) = \frac{4}{3}\pi r^3$. Find $V'(5)$.

6. Demand is $D(p) = 1200 - 30p$. Interpret $D'(p)$.

7. If $Q(t) = 500e^{-0.08t}$, find the percent rate of change.

8. A cube has side $x(t)$. Write the rate of change of volume in terms of x and x'.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. A rectangular region has $A(t) = L(t)W(t)$. If $L = 8$, $W = 5$, $L' = 0.3$, and $W' = -0.2$, find A'.

10. For $C(q) = 1000 + 12q + 0.005q^2$, compare average cost C/q and marginal cost at $q = 500$.

11. A density $\rho = m/V$ changes with time. Derive ρ'.

12. Explain why $A'(r) = 2\pi r$ for circle area has the same numerical form as circumference.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. A company's revenue is $R(q) = q(120 - 0.02q)$. Find output levels where marginal revenue is positive.

14. A sphere's radius grows at a rate proportional to $1/r^2$. Show that its volume grows at a constant rate.

15. If $Q'(t)/Q(t) = k$ is constant and $Q > 0$, identify the model form.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. A quantity Q has units joules and depends on temperature T in kelvins. What are the units and meaning of d^2Q/dT^2?

17. Suppose $R(p) = pD(p)$ and demand decreases with price. Derive a condition for revenue to increase with price.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. If $A(r)$ is area in m^2 and r is meters, units of $A'(r)$ are

(A) m^3 **(B)** m^2
(C) m **(D)** 1/m

2. Marginal cost $C'(q)$ is best interpreted as

(A) total cost **(B)** average cost per unit
(C) approximate change in cost for one more unit **(D)** profit per unit

3. For $Q(t) = Ae^{kt}$, the relative rate Q'/Q equals

(A) A **(B)** k
(C) kt **(D)** e^{kt}

4. If $L' > 0$ and $W' < 0$, the sign of $(LW)'$

(A) must be positive **(B)** must be negative
(C) must be zero **(D)** depends on the values and rates

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $P(q) = 80q - 0.03q^2 - 4000$, marginal profit at $q = 900$ is

(A) 16 **(B)** 26
(C) 53 **(D)** 80

6. If $N(t) = 2500(1.04)^t$, the instantaneous percent growth rate is closest to

(A) 3.92% **(B)** 4.00%
(C) 4.08% **(D)** 4.16%

AP-Style Free-Response Practice

No Calculator

A company's daily cost is $C(q) = 6000 + 18q + 0.012q^2$ dollars when q items are produced.

(a) Find and interpret $C'(400)$. [2 points]

(b) Find average cost at $q = 400$. [2 points]

(c) Explain why marginal cost and average cost need not agree. [2 points]

Calculator Permitted

The electrical power use $W(t)$ is represented by the shared table.

(a) Estimate the average rate of change of W on $[2, 6]$. [2 points]

(b) Compare the average rate with $W'(4)$ and explain. [2 points]

(c) Use $W(6)$ and $W'(6)$ to estimate $W(6.5)$. [2 points]